📝 BUS 4101: E-Commerce Class Test Quick Review

Focus: Classes 01–03 | JUST Dept. of CSE

1. Core Definitions & Distinctions

- **E-Commerce:** The process of buying/selling goods, services, and info over electronic networks (primarily the Internet). Focus is on **transactions**.
- **E-Business:** The broader operational use of digital technology to optimize *all* business processes (Internal ERP, HR, etc.).
  - *Exam Tip:* **E-Commerce is a subset of E-Business.** All E-Commerce is E-Business, but not the other way around.

2. The 8 Unique Features of E-Commerce

Memorize these definitions and examples for MCQ/Short Answer:

1. **Ubiquity:** Available everywhere, 24/7 (e.g., shopping on Daraz at 2 AM).
2. **Global Reach:** Crosses boundaries (e.g., selling BD handloom to Germany via Etsy).
3. **Universal Standards:** Shared technical standards (HTTP/HTML) reduce entry costs.
4. **Richness:** Complex content (video/audio/animation) improves product messaging.
5. **Interactivity:** Two-way communication (Chatbots/Product configurators).
6. **Information Density:** More info available to all (Price comparison sites like Shopper.com.bd).
7. **Personalization/Customization:** Targeting specific users (Amazon's "Customers who bought this...").
8. **Social Technology:** User-generated content/sharing (Facebook Marketplace/Instagram Shops).

3. Revenue Models (How firms make money)

Model	Mechanism	Examples
Advertising	Charging to reach a large user base	Google Ads, Facebook, YouTube
Subscription	Fixed recurring payments	Netflix, Spotify, LinkedIn Premium
Transaction Fee	Commission per sale	**Daraz (3-10%)**, Upwork, bKash

Freemium	Free base + paid premium	Zoom, Dropbox, Spotify Free/Premium
Affiliate	Commission for referral traffic	Amazon Associates, Daraz Affiliate
Sales (E-tail)	Direct product margin	**Chaldal**, Apple.com

4. Industry Dynamics & Growth Laws

- **Metcalfe's Law:** The value of a network is proportional to the square of the number of users ($V \propto n^2$).
  - *Calculation Tip:* If users double ($2x$), value quadruples ($4x$).
- **Disintermediation:** Removing the middleman (e.g., Artists selling directly on Spotify).
- **Re-intermediation:** Creating *new* digital middlemen (e.g., Booking.com replacing travel agents).
- **Network Effects:** Service becomes more valuable as more people join (Direct: WhatsApp; Two-sided: Uber riders/drivers).
- **The Long Tail:** Aggregating demand for niche products that wouldn't survive in physical stores.
- **Winner-Takes-All:** Network effects lead to dominant players (Google 93% Search, Daraz dominant in BD).

5. Value Chain (Porter's Model)

- **Primary Activities:** Inbound Logistics → Operations → Outbound Logistics → Marketing & Sales → Service.
- **Support Activities:** Infrastructure, HR, Technology Development (crucial for E-commerce), and Procurement.

6. Business Models (Participants)

- **B2C:** Business to Consumer (Daraz, Netflix).
- **B2B:** Business to Business (Alibaba, Walmart's Retail Link, **e-GP Portal BD**). *Largest volume globally.*
- **C2C:** Consumer to Consumer (OLX, eBay, Facebook Marketplace). Challenge: **Trust**.
- **C2B:** Consumer to Business (Upwork/Fiverr freelancers, Influencers).
- **G2C/B2G:** Government models (NID services, e-Tax filing, e-GP portal).

💡 Final Minute Tips

- **Bangladesh Context:** Always mention **bKash** (MFS), **Daraz/Chaldal** (B2C), and **e-GP** (B2G) in your descriptive answers.
- **Amazon Case:** Remember Amazon uses *multi-sided* revenue models: Retail (Sales), Marketplace (Transaction fees), Prime (Subscription), and AWS (Infrastructure).

📝 BUS 4101: E-Commerce Class Test (Architecture, Marketing & Logistics)

Focus: Classes 04–06 | JUST Dept. of CSE

1. Class 04: Architecture & Web Technologies

Three-Tier Architecture

- **Tier 1: Presentation Layer (Front-End):** What users see (HTML/CSS/JS, React).
- **Tier 2: Business Logic Layer (Application Server):** The "brain" (Python, Java, Node.js). Handles rules, orders, and auth.
- **Tier 3: Data Layer (Database):** Persistent storage (MySQL, MongoDB). Isolated for security.

REST vs. SOAP

Feature	REST	SOAP
Format	JSON (compact/fast)	XML (verbose/heavy)
Protocol	Architectural Style	Strict Protocol
Use Case	Public APIs, Mobile apps	Banking, Legacy systems

Scalability & Performance

- **Horizontal Scaling (Scale Out):** Adding more machines (e.g., Daraz 11.11 spike). Most cost-effective.
- **Vertical Scaling (Scale Up):** Adding power to one machine (RAM/CPU). Has physical limits.
- **CDN (Content Delivery Network):** Geographically distributed servers that cache static content (images/CSS) to reduce **latency**.

2. Class 05: Digital Marketing & Analytics

The AIDA Funnel

1. **Awareness:** SEO, Social Ads (Goal: Reach).
2. **Interest:** Blogs, Newsletters (Goal: Educate).

3. **Desire:** Reviews, Testimonials, Live Chat (Goal: Preference).
4. **Action:** Optimized Checkout, Trust Badges (Goal: Close Sale).

Key Formulas (High Exam Probability)

- Conversion Rate (CR): $CR = \frac{\text{Purchases}}{\text{Visits}} \times 100$
- Customer Acquisition Cost (CAC): $CAC = \frac{\text{Total Marketing Spend}}{\text{New Customers Acquired}}$
- Average Order Value (AOV): $AOV = \frac{\text{Total Revenue}}{\text{Number of Orders}}$
- Customer Lifetime Value (CLV): $CLV = AOV \times \text{Purchase Frequency} \times \text{Customer Lifespan}$
- Return on Ad Spend (ROAS): $ROAS = \frac{\text{Revenue from Ads}}{\text{Ad Spend}}$
- Metcalfe's Law: $V \propto n^2$

3. Class 06: Logistics & Supply Chain

Fulfillment Models

- **Dropshipping:** No inventory held; supplier ships directly to customer. Low risk, lower margins.
- **3PL (Third-Party Logistics):** Outsourcing warehousing to specialists (e.g., Amazon FBA, Pathao Fulfillment).
- **4PL (Fourth-Party Logistics):** An integrator that manages multiple 3PLs (e.g., **Alibaba's Cainiao Network**).

Inventory & EOQ

The **Economic Order Quantity (EOQ)** minimizes total cost (Ordering + Holding costs).

$$EOQ = \sqrt{\frac{2DS}{H}}$$

- D: Annual Demand
- S: Ordering Cost (per order)
- H: Holding Cost (per unit per year)

Last-Mile Challenges (Bangladesh Context)

- **Address Quality:** Non-standardized/descriptive addresses.
- **Traffic:** Dhaka congestion increases costs and reduces driver capacity.

- **Disaggregation:** High cost of splitting bulk shipments into individual home deliveries (53% of total shipping cost).

💡 Quick Case Studies

- **SSL Commerz:** Leading BD payment aggregator using a REST API.
- **Cainiao:** Alibaba's "asset-light, data-heavy" 4PL model that orchestrates 3rd party carriers via APIs.
- **A/B Testing:** Controlled experiment between Version A (Control) and B (Variant) to improve Conversion Rate.

📝 BUS 4101: E-Commerce Class Test (Payments, Security & Law)

Focus: Classes 07–09 | JUST Dept. of CSE

1. Class 07: Payment Systems

The Card Ecosystem (Roles)

- **Issuing Bank:** The cardholder's bank (e.g., DBBL, BRAC). Authorizes/declines transactions.
- **Acquiring Bank:** The merchant's bank (e.g., EBL). Maintains the merchant account.
- **Payment Gateway:** The tech bridge (e.g., **SSL Commerz**). Encrypts and transmits data.
- **Card Network:** The "rails" (Visa, Mastercard). Routes messages between banks.

Key Concepts

- **Authorization:** Real-time check for funds. **Does not move money**; only reserves it.
- **Settlement:** The actual transfer of funds (usually $T + 1$ to $T + 3$ days).
- **Tokenization:** Replacing the 16-digit card number (PAN) with a random "token" to prevent data theft.
- **MFS (Mobile Financial Services): bKash** and **Nagad**. Essential for the unbanked in BD.

2. Class 08: Encryption & Security Protocols

The 5 Dimensions of Security

1. **Confidentiality:** Only authorized people can read (Threat: Interception).
2. **Integrity:** Data hasn't been changed (Threat: Tampering).
3. **Availability:** System is up (Threat: DDoS).
4. **Authenticity:** Identity is genuine (Threat: Phishing).
5. **Non-repudiation:** Cannot deny a transaction (Threat: Order dispute).

Encryption & Hashing

- **Symmetric (AES):** One key for both. Fast, used for bulk data.
- **Asymmetric (RSA):** Two keys (Public/Private). Used for key exchange and signatures.
  - **RSA Formula (High Probability):** $C = M^e \pmod{n}$ (Encryption).
- **Hash Function (SHA-256):** One-way, fixed-size digest. Used for passwords and integrity.

- **Digital Signatures:** Rule: **Sign with Private Key, Verify with Public Key.**

SSL/TLS Handshake

- **Purpose:** Authenticate the server and establish a secret symmetric **session key**.
- **CA (Certificate Authority):** Trusted third party (e.g., Let's Encrypt) that issues digital certificates.

3. Class 09: Threats, Ethics & Legal Issues

Technical Threats

- **SQL Injection (SQLi):** Malicious SQL in input fields. Prevention: **Parameterized queries**.
- **XSS (Cross-Site Scripting):** Injecting JS into pages. Goal: Steal session cookies.
- **CSRF:** Tricking a logged-in user's browser into performing actions. Prevention: **CSRF Tokens**.
- **Social Engineering:** Phishing (generic), Spear Phishing (targeted), Vishing (voice), Smishing (SMS).

Ethics & Dark Patterns

- **Dark Patterns:** UI tricks (e.g., "Confirmshaming," hidden costs/drip pricing, "Roach motels" for hard cancellation).
- **Price Discrimination:** Charging different prices based on willingness to pay (e.g., iOS vs. Android users).

Privacy & Law

- **GDPR (EU):** Opt-in consent, Right to be Forgotten, 72-hour breach notice. (Fines: 4% of global revenue).
- **Digital Security Act (DSA) 2018 (BD):**
  - **Section 17:** Attacking Critical Infrastructure (14 years).
  - **Section 19:** Unauthorized Access/Hacking (7 years).
- **ICT Act 2006 (BD):** Gives **legal validity** to electronic documents and digital signatures.

💡 Quick Exam Checklist

- **RSA Rule:** What one key encrypts, *only* the other key can decrypt.
- **Hashing Rule:** You cannot reverse a hash to find the original password.